

SHRESTH JAISWAL

Bhopal, India | +91 8470980674 | jaishhresth143@gmail.com

[LinkedIn](#) | [GitHub](#) | [Portfolio](#) | [LeetCode](#)

EDUCATION

B.Tech in Computer Science and Engineering — Vellore Institute of Technology

Sep 2023 – Present

CGPA: 8.40 / 10.0

SKILLS

AI/ML & LLMs: LLM APIs (Groq, OpenAI GPT-4o, Gemini), Prompt Engineering, Conversational AI, LangChain, LangGraph, Agentic Workflows (Function Calling, Tool Use), Agent Testing

Voice AI: ElevenLabs (TTS), Web Speech API, Web Audio API, Twilio (Twiml webhooks)

Languages: Java, Python, JavaScript, SQL

Backend: FastAPI, Node.js, Express.js, REST APIs

Frontend: React.js, Redux Toolkit, React Hook Form, HTML, CSS, TailwindCSS

EXPERIENCE

Software Engineering Intern — Scramble Labs Technology Pvt. Ltd.

May 2025 – Aug 2025

- Built REST APIs in Python, Django and PostgreSQL with request validation and query optimization.
- Wrote unit and integration tests for the new API endpoints.
- Built React and TypeScript components in a production Next.js application and improved page load performance using component caching and memoized render hooks.

PROJECTS

EstateStream AI — Voice Agent Prototype for Real Estate Sales

Python, FastAPI, OpenAI GPT-4o, ElevenLabs, PostgreSQL, SQLAlchemy, Alembic, React, WebSockets, pytest | [Code](#) | [Live Demo](#)

- Built an inbound lead qualification agent in Python and FastAPI. GPT-4o function calling searches property listings, saves leads to a CRM table, and escalates to a human broker.
- Grounded property answers in database results across 22 seeded listings, so a search with no match returns nothing instead of an invented listing.
- Built a browser voice client over WebSockets with browser speech recognition, ElevenLabs text to speech, client side barge in, a Web Audio waveform, and a React dashboard with transcripts and automatic post call summary tags.
- Wrote 19 pytest tests, including 10 adversarial conversation cases (interruptions, silence, hostile callers, broker escalation), run against a rule based mock LLM for fast offline testing. Added Twilio Twiml routing, Alembic migrations, and Docker config.

Multi-Agent Research Assistant

LangGraph, Streamlit, Groq, Tavily, Python | [Code](#) | [Live Demo](#)

- Architected a multi-agent research assistant using LangGraph state machines with 5 specialized nodes including Router, Researcher, Orchestrator, Worker, and Reducer, enabling autonomous web research and information synthesis.
- Developed a 3-mode research pipeline (Closed Book, Hybrid, and Open Book) using Tavily Search API to search multiple sources and generate fact-based, citation-backed answers.
- Launched a stateless Streamlit application on Community Cloud using Groq API, featuring live graph execution streaming, real-time node progress tracking, and one-click Markdown export.

FiTFEAST — Multi-Agent AI Indian Meal Recommendation System

LangGraph, Groq API, Flask, Python | [Code](#) | [Live Demo](#)

- Architected a 4-agent LangGraph workflow (Medical Analyzer, Nutritionist, Meal Generator, Validator) with a 2-iteration validation loop, enabling autonomous reasoning for personalized 7-day Indian meal plans.
- Leveraged Groq/compound, prompt engineering, and PyPDF to analyze medical reports, enforce clinical constraints and allergen safety, and generate medically-aware meal recommendations.
- Built a Flask + Server-Sent Events (SSE) application with an HTML, Tailwind CSS, and JavaScript frontend, and deployed using Gunicorn for production-ready scalability.

ACHIEVEMENTS & LEADERSHIP

- Community Leadership: Core Member of the Notion Community at VIT Bhopal — organized tech talks and peer learning workshops.
- Media Recognition: Featured in local newspaper for scoring 98.6% in Class 10 boards, ranking in the top 3 in the district.